

Aligned but Inert: Concept Bottlenecks in Dense Prediction, and How to Make Them Causal Supplementary Material

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This supplement gives the implementation detail needed to reproduce the paper, the full version of every table the paper summarizes, and the results the paper mentions but had no room to tabulate—including the two where our own method loses. Section references of the form “Sec. 4” point into the main paper; “Sec. S1” points here.

S1. Implementation and reproducibility

Data. BraTS 2024 adult glioma. Four co-registered modalities (T1n, T1c, T2w, T2f), skull-stripped and interpolated to 1 mm^3 by the challenge organizers. Native labels $\{0, 2, 4\}$ are remapped to $\{0:\text{Background}, 1:\text{NCR}, 2:\text{Edema}, 3:\text{ET}\}$. Each modality is z -scored over its non-zero voxels, then a 128^3 patch is taken. The split is patient-wise 75/12.5/12.5 with seed 42, giving $n_{\text{test}}=168$. It is our own split, so absolute Dice is not comparable to BraTS leaderboard results; every comparison in the paper is within-split.

Backbones. Two, both from MONAI. *SegResNet* is the headline backbone and carries every configuration comparison. *Swin* + UNETR decoder is the backbone on which the failure was first diagnosed; its encoder is initialized from MONAI’s self-supervised `model.swinvit.pt`, which requires three key remappings (strip the `module.` prefix from `DataParallel`, rename `mlp.fc1/fc2` to `linear1/linear2` for MONAI 1.5, and inflate `patch_embed.proj.weight` from 1 to 4 input channels). All 126 encoder keys load.

Module. $K=4$ concepts, bottleneck 192×16^3 (4096 tokens of width 192). The router is linear plus K learnable prototypes with a learnable temperature; experts are pre-norm residual MLPs. Training uses soft dispatch, inference hard dispatch.

Table S1. Three-phase curriculum. λ_1 is zero during warm-up, which is the default we inherited; ablation A4 (Sec. S6) shows it is not actually required.

phase	epochs	λ_1	λ_2	λ_3	λ_4	τ
warm-up	1–10	0.0	0.5	0.00	0.0	2.0
alignment	11–60	1.0	0.5	0.01	0.3	1.0
refinement	61–80	1.0	0.1	0.01	1.0	0.5

Table S2. What separates the configurations. Nothing else differs.

	expert	skip gate α	additive
NATURAL	$h + c_k(h)$	1.0	no
GATED	$c_k(h)$	0.5	no
DIRECT	$h + c_k(h)$	1.0	yes, $\lambda=1$

Optimization. AdamW, learning rate 10^{-4} with cosine decay, weight decay 10^{-5} , gradient clip 1.0, mixed precision, 80 epochs. Batch size 1 with 4 gradient-accumulation steps (effective batch 4); gradient checkpointing is on, which is what makes the model fit in 16 GB. Runs were performed on a single T4 and a single L4.

Curriculum and loss weights. $\mathcal{L} = \mathcal{L}_{\text{seg}} + \lambda_1 \mathcal{L}_{\text{align}} + \lambda_2 \mathcal{L}_{\text{div}} + \lambda_3 \mathcal{L}_{\text{ent}} + \lambda_4 \mathcal{L}_{\text{bnd}}$, with $(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ set by phase (Table S1). $\mathcal{L}_{\text{align}}$ is a foreground-only focal BCE with $\gamma=1.0$ and per-concept weights $[1, 3, 1, 1]$, which up-weights NCR because it is by far the scarcest concept at 16^3 ; an additional contrastive term of weight 0.3 separates NCR from ET, the pair the router confuses most. λ_3 is capped at 0.01: larger values push routing toward uniform and destroy CAS_{fg} .

The three configurations. All three share backbone, losses, curriculum, optimizer, split and seed, and differ only in the switches below, so every comparison isolates the architectural change.

Table S3. Complete label-flip matrices ($n=168$). Rows are the source concept, columns the target; each entry is the fraction of re-routed voxels whose predicted label became the target. Diagonals are the identity control. The paper quotes the mean of the six off-diagonal cells under the tumor-restricted scope.

from \ to	NCR	Edema	ET
<i>GATED, tumor-restricted</i> — mean 0.011			
NCR	0.000	0.028	0.055
Edema	0.007	0.000	0.004
ET	0.013	0.009	0.000
<i>DIRECT, tumor-restricted</i> — mean 0.636			
NCR	0.000	0.312	0.280
Edema	0.661	0.000	0.772
ET	0.515	0.738	0.000
<i>DIRECT, all tokens</i> — mean 0.816 reported for completeness; see the scope discussion, Sec. 5.4			

Table S4. Restricting the scope lowers the headline and improves both controls (DIRECT, $n=168$, mean over off-diagonal pairs).

	tumor-restricted	all tokens
target class $\uparrow$	0.540	0.798
source class $\downarrow$	−0.544	−0.514
off-target $\rightarrow 0$	0.079	0.151
outside region $\rightarrow 0$	0.005	0.016

Table S5. Expert statistics ($n=40$). Residual scale is $\|E_k(h) - h\|/\|h\|$; divergence is $\|E_a(h) - E_b(h)\|/\|h\|$ averaged over pairs. NATURAL is measured on Swin ($\dagger$), the backbone on which the design was first diagnosed.

	residual scale	pairwise divergence
NATURAL $\dagger$	0.59–0.74	0.93
GATED	≈ 1.01	0.16
DIRECT	0.21–0.47	0.42

Where each number comes from. Scripts are `pipeline/routing_intervention.py` (Test 1), `pipeline/expert_diagnostics.py` (Test 2 and expert statistics), `pipeline/linear_probe.py` (Test 3), `pipeline/evaluate.py` (Dice, CAS_{fg}, AU-ROC, ECE) and `pipeline/error_detection.py` (Sec. S7). Each writes a per-case CSV and a summary JSON under `evals/`; the tables here are rendered from those files, so the paper’s figures and tables cannot drift apart.

S2. The intervention, in full

The paper reports the mean off-diagonal label-flip. Table S3 gives every cell, for both configurations and both scopes. Two things are visible here that a single mean hides.

First, the diagonal is *exactly* zero in all four matrices. This is the identity control: forcing the tokens already assigned to a through a ’s own expert is a bit-exact no-op, so every off-diagonal entry is attributable to re-routing rather than to the instrumentation. Achieving this required hardening the router’s logits to a one-hot for *every* token rather than only the intervened ones (Sec. 3.4); without that, an identity intervention moves the prediction by 14.3 logits and the control is worthless.

Second, the asymmetry the paper discusses is a property of rows, not cells. NCR is a weak source (0.28–0.31) and a decent target (0.51–0.66) under DIRECT, and the effect size tracks how confidently the model held the original concept.

Why we quote the smaller number. A router sends most of the volume to background, and roughly a quarter of all tokens to the edema expert—most of them normal tissue. Re-routing every token of a concept therefore repaints the whole brain and reports 0.816. That figure is real but close to tautological under additive routing: it shows the output

follows the routing everywhere. The tumor-restricted scope measures concept control where it is clinically meaningful, and it is also *cleaner* on both controls, not merely smaller (Table S4).

S3. Expert diagnostics

The paper rules out expert degeneracy as the explanation for inertness. Table S5 gives the underlying numbers for all three configurations, and they carry a result that is easy to miss: the subtractive repair did not merely fail, it made the experts *interchangeable*. Removing the residual so that $E_k(h) = c_k(h)$ drops mean pairwise divergence from 0.93 to 0.16. With a residual, each expert supplies only a concept-specific correction and is free to differ; without one, every expert must independently reconstruct whatever the decoder needs, so they converge on nearly the same map. This is the clearest single piece of evidence for the paper’s claim that subtractive repairs backfire.

Ablating a two-path module. Under additive routing the module reaches the output twice, and each path must be ablated separately. Table S6 gives all four conditions. *by-pass* replaces the module output with raw encoder features; *zero_expert* zeros the features the decoder consumes; *zero_routing* zeros the additive term; *zero* removes both. Patching only the decoder’s bottleneck input—the obvious thing to do—leaves the additive term intact and measures a 0.09 logit change where the true effect is 7.0. A study using that measurement would conclude the bottleneck is unnecessary and reject a working design.

An interaction we cannot explain. Zeroing *both* paths is less damaging than zeroing the routing term alone (edema

Table S6. Bottleneck necessity for DIRECT (per-class Dice, $n=40$). Removing the routing term alone destroys the prediction; this is the same model whose inert predecessor lost at most 3.1 Dice to having its entire bottleneck zeroed.

	base	bypass	zero_expert	zero_routing
NCR	0.483	0.484	0.053	0.006
Edema	0.898	0.477	0.333	0.418
ET	0.852	0.597	0.428	0.172

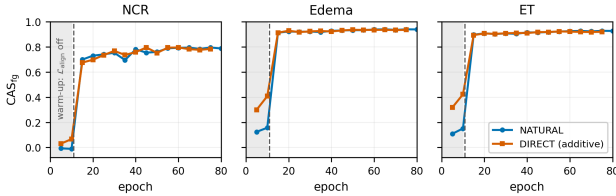


Figure S1. CAS_{fg} during training. The shaded band is warm-up, where $\mathcal{L}_{align}$ is off. NATURAL's routing sits near the floor until alignment supervision switches on at epoch 11; DIRECT's climbs during warm-up, because the routing logits are part of the output and the segmentation loss therefore trains them.

0.801 vs. 0.418). Our best guess is that the decoder, trained to add to a routing base, is left miscalibrated when the base is stripped but the expert path remains, whereas zeroing both puts it in a cleaner regime. We report it because it is reproducible and we do not have a confirmed account of it.

S4. Alignment emerges from the segmentation loss

A natural objection to CAS_{fg} is that it only measures what $\mathcal{L}_{align}$ trains. Additive routing answers it directly. Because the routing logits are a term of the output, $\mathcal{L}_{seg}$ alone trains them to be concept-aligned—so alignment should appear during warm-up, when $\mathcal{L}_{align}$ is switched off entirely. It does (Fig. S1). At epoch 10, with λ_1 still zero, DIRECT has reached CAS_{fg} 0.41 on edema and 0.42 on ET against 0.16 and 0.15 for NATURAL. The alignment is a consequence of the architecture, not only of the supervision.

S5. Post-hoc baselines: the comparison we lose

The paper concedes that post-hoc attribution wins on deletion and insertion AUC and does not tabulate it. Table S7 gives the numbers. Integrated gradients and occlusion reach ≈ 0.95 deletion, far above what routing achieves.

We read this as a statement about what the metric measures rather than a defeat. Deletion and insertion score *input sensitivity*: how fast the prediction degrades as highly-ranked voxels are removed. A method that is optimized to find sensitive inputs will win, and integrated gradients is ex-

Table S7. Deletion and insertion AUC against ground-truth regions (Swin, $n=168$; NCR $n=67$, ET $n=163$ where the concept is present). Higher is better for both. Post-hoc attribution wins this comparison and we report it in full.

	NCR		Edema		ET	
	del	ins	del	ins	del	ins
Occlusion	0.952	0.972	0.942	0.897	0.961	0.918
Int. grad.	0.962	0.901	0.948	0.898	0.946	0.929
Gradient	0.787	0.461	0.674	0.475	0.460	0.411
Grad-CAM	0.455	0.350	0.392	0.369	0.400	0.356

actly that. It is not a measure of whether the model's own concept assignment is load-bearing, which is the question this paper asks and which no saliency map can be asked at all—there is nothing in a readout to intervene on.

Concept-discriminability, per concept. The paper reports the mean. Per concept, and taking both sides from the same run (Swin[†], foreground AUROC against other tumor voxels), the routing reaches 0.854/0.891/0.872 on NCR/edema/ET against 0.850/0.778/0.888 for a linear probe on the features entering the module. The probe is competitive on NCR and marginally *better* on ET; the concept machinery earns its +0.033 mean advantage almost entirely on edema. For reference, SegResNet's routing reaches 0.853/0.905/0.889, but we do not have a matched probe for it and so do not form that comparison.

S6. Ablations

A1 — no alignment loss. Setting $\lambda_1=0$ throughout collapses CAS_{fg} to $-0.011/0.322/0.094$ and concept-discriminability to $0.294/0.843/0.274$, while Dice barely moves (WT 0.905 vs. 0.927). Alignment is therefore *learned*, not incidental to competent segmentation. The same number carries the paper's uncomfortable corollary: the segmentation does not need the alignment.

A4 — no warm-up. Switching $\mathcal{L}_{align}$ on from epoch 1 was expected to cause expert collapse. It does not: CAS_{fg} reaches $0.666/0.932/0.927$ against $0.688/0.935/0.930$ for the full curriculum, and Dice WT is 0.908. With a pre-trained encoder, prototype-anchored router and a diversity term active from the first epoch, warm-up is redundant. Its only measurable effect is a mild improvement in utilization balance. We keep it as a safe default and report this as a robustness result rather than a justification.

S7. Uncertainty: a negative result

Routing entropy is a natural per-voxel uncertainty signal and we tested whether it predicts where the model is wrong. It does not, and plain segmentation confidence—available

Table S8. Predicting voxel-level model error (AUROC, $n=168$, mean error rate in region 0.171). The best routing-derived signal is 0.097 *below* plain segmentation confidence.

signal	error-detection AUROC
segmentation confidence	0.883
routing/decoder disagreement (soft)	0.786
routing/decoder disagreement (hard)	0.726
routing entropy	0.560

Table S9. The natural design on both backbones ($n=168$). Dice is over nested BraTS regions; CAS_{fg} and discriminability are per concept.

	Swin + UNETR	SegResNet
Dice WT/TC/ET	0.915/0.845/0.837	0.927/0.883/0.869
CAS _{fg} N/E/T	0.69/0.94/0.93	0.80/0.96/0.95
concept-discrim.	0.872	0.883
causally inert	yes	yes

but that expectation is untested outside the two backbones here.

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from the same forward pass, with no concept machinery—beats every routing-derived signal we tried (Table S8). Routing ECE is 0.65–0.84 across configurations, which is poorly calibrated in absolute terms. We therefore make no uncertainty claim anywhere in the paper and do not position routing entropy against MC-dropout or deep ensembles.

S8. Backbone transfer

Table S9 gives both backbones on the natural design. They differ in pretraining, capacity and accuracy, and agree on everything that matters to the paper’s argument: both align well, and both are inert. The transformer additionally shows the placement failure in its sharpest form—its module sits at stage 2 of 5, so the two deeper and more semantic encoder stages bypass it entirely, which is worth stating plainly because our own first design made exactly that error while the paper it produced said “encoder bottleneck”.

S9. Limitations restated

For convenience, the limitations declared across the main paper, collected. One dataset and one anatomy. No reader study, so no claim that clinicians want concept-level explanations. No plain-backbone accuracy control, so the ≈ 1 Dice cost is a relative figure against the same architecture with an inert bottleneck, not the cost of adding the module to a plain network. The NATURAL causal numbers are a 5-case pilot on the Swin backbone; GATED carries $n=168$ on SegResNet and is equally inert. A 16^3 bottleneck makes the explanation coarse, and NCR—tens of tokens per case—is the consistent weak spot across every metric we report. $K=4$ follows the BraTS protocol and forces transition-zone tokens into a single concept. Finally, the two leaks are properties of an architecture family rather than of this dataset,